

STATEMENT OF WORK

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1. Introduction/Background:

The NASA Aeronautics Research Mission Directorate's (ARMD) Innovative Concepts (IC) team identifies aeronautics focus areas that have potentially transformative opportunities that can be further matured for NASA.

As part of the work, multi-faceted data from wide and varied sources are ingested to identify potential focus areas for further exploration. This data may include examining emerging markets and understanding societal, technological, and industrial ecosystems and trends. MADI represents a strategic response within NASA to the growing need for secure, scalable artificial intelligence solutions within the federal government. It combines enterprise-grade security, modular extensibility, data visualization, and integration of custom plug-ins and data (e.g., all NASA TechPort) to support advanced discovery human-AI collaboration capabilities. These help researchers to identify connections between disparate research areas and highlight potential white spaces across NASA, among and other agency's, research portfolio. Sprints may need different types of facilitation methods in the fields of strategic foresight and design.

Technical Infrastructure

MADI's system architecture comprises a Node.js and Python backend, PostgreSQL database implementation, and React-based frontend, deployed via serverless services in Google Cloud Run. The system currently operates through NASA's Cloud Compute Services Google Cloud Platform (GCP) instance, maintaining compliance with federal security protocols through NAMS authentication systems and transitioning to a Model Garden hosted within GCP or Azure's government cloud infrastructure to ensure FISMA compliance. The system's codebase adheres to open-source licensing standards (MIT, ISC, BYD-3) for all components except cloud-native services and AI models, facilitating potential adoption by other federal agencies with similar authorization to operate (ATO) requirements.

1.0 Scope of Work:

This work supports the lifecycle of foresight within the Innovative Concepts (IC) team in the Transformative Tools and Technologies (TTT) project, within the Transformative Aeronautics Concepts Program (TACP) of ARMD. This includes data collection, analysis, synthesis, dissemination and knowledge management. It also supports stakeholders from ARMD and elsewhere in the agency, as identified by the IC lead.

- 1.1 Market research around societal needs, trends and capabilities

- 1.2 Integration of Social listening, Patent and academic literature scraping, company information into MADI to enable visualization and search of 'whitespace'

2.0 Objectives:

1. To understand the current state of technology and trend landscape that potentially supports an identified focus area. This includes specific technologies, challenges, development paths and timeframes.
2. To understand emerging technology, adoption, use trends within a discipline (e.g. energy storage or gaming platforms).
3. To understand key players (investors, researchers, users etc.) in the area.
4. To understand methods/processes and best practices to improve organizational agility in order to allow the organization to enable faster decision making and quicker responses unseen challenges
5. To understand what stakeholders in the opportunity areas see as major challenges or needs

3.0 Description of the Work/Contractor Tasks:

1. Provide AI Consumer Insight (AICI) analysis
 - a. Leveraging PreScouter's expertise in data analytics and sentiment analysis in order to provide insights regarding consumer sentiments towards different technologies, ideologies, and organizations. Potential platforms to be analyzed may include:
 - i. Reddit
 - ii. Ground News
 - iii. Youtube comments
 - b. Deliver every 3 months
2. Updates to previous PreScouter NASA work
 - a. Belief systems analysis
 - b. Airport door to door PAX investigation
 - c. Creative one-off experiments/news/ implications
3. Helping NASA see trends more broadly across social, political, technical landscapes to understand new opportunities and where things are moving
 - a. Insights from PreScouter's work across nearly every industry, to be customized in terms of frequency and methodology to fit NASA's needs technologies, industries, topics of interest, organizations, etc.
 - b. Periodic trend reports across different domains
 - c. Every 6 months
4. Topic Briefings
 - a. Based on IC team workstreams

NASA will provide the following:

- Suggested problem areas informed by NASA strategic guidance;
- NASA personnel to serve as SME for market and technology knowledge elicitation (non-exhaustive)
- Virtual meeting environment for monthly meetings and any intermediate discussions that are needed.

Period of Performance

The contractor shall be available during the base period of 12 months for consultation and iteration (within reason). The period of performance will start immediately following task order award.

4.0 Deliverables:

1. AICI Monitoring License - 12 months of AICI monitoring. PreScouter to provide advanced data analytics and sentiment analysis to gain deep insights into broad perspectives on societal challenges, pain points and how these challenges could be overcome by new developments. The analytics platform we will support over the term of this license will support interrogation of the raw data and trends along multiple axis including topics of interest, demography, sources, information-bias, and engagement among other factors. This platform will also support integration with NASA's AI toolset. The specific methods and processes used to synthesize the data, visualize the data are provided by the NASA team while implementation is provided by PreScouter. The framework in which the data is being examined is also provided by NASA.
2. Ten (10) Intelligence Briefs – License for investigating and tracking novel events, technologies, methodologies, and concepts to assist NASA with its internal R&D, strategy, and engineering efforts. PreScouter will meet with NASA over the course of a series of Intelligence Brief meetings to address research objectives. PreScouter will leverage NASA internal briefing documents in order to align research with NASA internal goals.